

Yujung Hwang

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RESEARCH INTERESTS

Artificial Intelligence (AI), Machine Learning (ML), Healthcare Application, Large Language Model (LLM), User Experience(UX)

EDUCATION

University at Albany, SUNY

Ph.D. in Information Science

M.S. in Artificial Intelligence for Business

Albany, New York, US

Sep. 2024 – Present

Conferred 2027

Seoul National University of Science and Technology (SEOULTECH)

Bachelor of Information Technology Management (ITM)

Seoul, Korea

Mar. 2020 – Feb. 2024

- GPA: 4.33/4.5 (98.1/100), Graduated with a Summa Cum Laude

Northumbria University (NU)

Bachelor of Information Technology Management (dual degree)

Newcastle, England, UK

Feb. 2021 – Feb. 2024

- GPA: First-class Honours

University of Hradec Králové

Exchange Student, Department of Information Management

Hradec Králové, Czech Republic

Jul. 2023 – Aug. 2023

PUBLICATION

Hwang, Y., Goel, S. (2026). "NeuroCurriculum: Translating Neurodiversity Theory into Adaptive Curriculum Design", Decision sciences journal of innovative education (Under Review).

Hwang, Y., Goel, S. (2026). "XR-Based Learning for ADHD: A Systematic Review of Adaptive Interaction Design", AMCIS 2026 (Accepted).

Subasi, A., **Hwang, Y.**, et al. (2026). "Generative AI, Large Language Models, Graph Neural Networks and Knowledge Graphs Applications", *Elsevier-Elsa* (forthcoming book chapter).

Hwang, Y., Kurt, F., Curebal, F., Keskin, O., & Subasi, A. (2025). "ContextualGraph-LLM: A Multimodal Framework for Enhanced Darknet Traffic Analysis." *Computers & Security*.

Hwang, Y., Kim, C., Park, G. M., and Kwon, H. Y. (2025) "Affective Audio Feature-based Multimodal Learning Model for Extracting Video", *Applied Soft Computing Journal*.

Hwang, Y., Kim, C., and Kwon, H. Y. "Emotion-based Multimodal Learning Model for Video Highlight Detection," *Proceedings of the Korean Information Science Society Conferences*, (), 2,151-2,153 (in Korean).

Park, J., Kwon, T., Kim, B., **Hwang, Y.**, and Sim, M. K. "Multi-Agent Reinforcement Learning Based Optimal PV-ESS Control in Grid," *2023 IEEE IAS Global Conference on Renewable Energy and Hydrogen Technologies (GlobConHT)*, pp. 1-6, Mar. 2023.

Kim, B., **Hwang, Y.** and Sim, M. "Optimizing Supply Chain Management Using Reinforcement Learning," *Proceedings of the Korean Information Science Society Conferences*, (), pp. 687-689 (in Korean).

RESEARCH EXPERIENCE

SUNY-ALBANY AI and Cyber Security Lab

Researcher / Supervisor: Prof. Sanjay Goel

Albany, New York, US

Aug. 2025 – Present

- Funded Research (New York State - Office for People With Developmental Disabilities): Working on AI-driven assistive technologies for neurodiverse populations.
- Research focus on adaptive learning, accessibility, and user-centered design for inclusive digital systems, leveraging AI.

SUNY-ALBANY AI4HEALTH Lab

Researcher / Supervisor: Prof. Abdulhamit Subasi

Albany, New York, US

Sep. 2024 – Present

- Conducting research on multimodal methods for cancer prediction by integrating histocartography with Knowledge Graph (KG) modeling and LLMs.
- Developed a novel LLM model that combines graph-based and textual features, achieving enhanced performance for Darknet cyber network intrusion detection.

Kean University GigWork Analytics Project

New Jersey, United States

Graduate Research Assistant / Supervisor: Prof. Ngoc Dao

Dec. 2024 – Jan. 2025

- Conducted research funded by the National Science Foundation and the Social Security Administration to predict gig workers (e.g., freelancers, Uber drivers) and analyze the impact of gig work on welfare benefit eligibility.
- Cleaned and prepared datasets through EDA, feature engineering, and missing data handling to develop machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and Neural Networks.
- Optimized model performance using hyperparameter tuning and k-fold cross-validation, delivering actionable insights into the socioeconomic implications of gig work.

SEOULTECH Applied Probability Lab

Seoul, Korea

Researcher / Supervisor: Prof. Sim, Min Kyu

Dec. 2021 – Dec. 2022

- Researched the optimization of energy system operations using Reinforcement Learning (RL) techniques.
- Assisted in research on voltage control mechanisms for force-commutated converters and applied a control algorithm using Proximal Policy Optimization.
- Conducted weekly seminars to review the latest RL papers, implementing new baseline code in Stable-Baselines3.

PATENT

Apparatus and method for multimodal based highlight detection in video using audio emotional information

- Authority: South Korea
- Application date: Jan 2024
- State: Under Prosecution

WORK EXPERIENCE

Newmix AI

Soeul, Korea

AI Engineer, Department of AI Product Development

May. 2025 – Aug. 2025

- Led a two-person team developing CSO-OCR, an AI-based web system using Clova OCR to extract drug and insurance data from handwritten prescriptions.
- Built an end-to-end OCR validation pipeline improving prescription data accuracy for pharmacy and insurance workflows, acquired by SS-Compare in Q1 2026.

Evident (Dental Workflow Automation)

Vancouver, Canada

AI Specialist, Department of Digital Marketing

Dec. 2024 – Aug. 2025

- Developed AI-powered analytics pipelines and LLM-based automation systems for real-time business intelligence and content workflows.

Nimbyx (Medical IT services and consulting)

Vancouver, Canada

UX Researcher, Department of Digital Agency

Jan. 2024 – Aug. 2024

- Capture requirements datapoints from key stakeholders and conduct user research for revamping Nimbyx's website.
- Create process flows, wire frames, prototypes, high fidelity mock-ups, and detailed UI and visual design specifications for web and mobile.

NexGen (Intelligent Software Development and Training)

Goyang, Korea

Front-end Developer and UX Engineer Intern, Department of IT Solution

Jun. 2023 – Sep. 2023

- Initiate a user-focused interface design with Adobe XD, fostering effortless navigation.
- Enhance a platform using HTML, CSS, and JavaScript, applying features such as user registration, PDF upload, and trading functionalities.

NeuralWorksLab Inc. (No-Code AI&ML Platform)

Seoul, Korea

UX Engineer and UI Design Intern, Department of IT Solution

Feb. 2023 – Jun. 2023

- Directed improvements in UX and brand recognition by creating easy-to-use workflows, attractive user interfaces, and distinctive web logos and icons for "Neural Studio" (a no-code AI tool) and "ChatSam" (an AI chat platform).
- Upgraded the functionality of the AI-based English education web "ChatSam" with a bespoke prompt engineering feature.
- Produced educational content in machine learning, using relevant datasets.

Hi-Media Computer Academy

Seoul, Korea

Computer Programming Instructor, Adult Education Team

Nov. 2021 – Jun. 2022

- Produced educational content in machine learning, using relevant datasets.
- Taught Python/Java/C/C++/Engineer Information Processing/Algorithm/HTML/CSS/JavaScript to college students.

HIGHLIGHTED PROJECT

Uni-Budget

Sep. 2022 – Feb. 2023

Team Project & Hackathon / Advisor: Prof. Jinwoo Jeong

- Developed "Uni-Budget," a Kotlin-based app employing OCR technology, to automate the Student Council's accounting audits by extracting data and comparing ledgers.
- Designed a robust architecture to efficiently transfer receipt image data between Flask and Android servers, improving the overall app's UX and functions.

Ping-Pong AI

Mar. 2022 – Dec. 2022

Start-up Club / Advisor: Prof. Sangheum Hwang

- Served as the president of a joint startup club of ITM-MSDE (Manufacturing Systems and Design Engineering) students.
- Led the development of a user-focused innovative service combining a ping-pong ball launcher with AI pose analysis software, completing Makerspace's manufacturing entrepreneurship training.
- Attracted US \$3,000+ in investment and acquired startup space through communications with venture capitalists.

TrendCatcher

Mar. 2021 – Jun. 2021

Team Project / Advisor: Prof. Hyukyeon Kwon

- Developed "TrendCatcher," a real-time web application leveraging Watchapedia Drama Rankings for industry predictions, incorporating Twitter data extraction via Natural Language Processing (NLP) and word cloud visualization.
- Architected a Hadoop-based storage and delivery system for bulk Twitter data and designed the overall web UI/UX, developing the website's front end.

AWARDS

1 st Place, Student Innovation Challenge: Innovate for Mental Health, University at Albany, SUNY	Oct. 2025
1 st Place, IT Service Development Contest, SEOULTECH	Dec. 2022
1 st Place, Global Challenger Contest, SEOULTECH	Dec. 2022
3 rd Place, AI Development Competition for Sustainable Energy, Korea Electronics Technology Institute	Sep. 2022
Grant Prize, Startup Development Competition, Ministry of Science and ICT	Sep. 2022
2 nd Place, Java Programming Competition, SEOULTECH	Feb. 2021
2 nd Place, University Student Online Education Donation Program Competition, Ministry of Education	Dec. 2020

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Soprano, MERRY Choir

Feb. 2021 – Dec. 2022

- Participated in choir performances for diverse audiences in locations such as subways and Gwanghwamun Square.
- Supported to the choir's mission of spreading happiness and creating positive impacts in communities.

Student Ambassador, SEOULTECH 'Aumi'

Nov. 2020 – Feb. 2023

- Administered and executed brand strategies, amplifying university exposure by creating YouTube and Instagram content, attending events for representation, and developing promotional materials.
- Strengthened stakeholder relations and raised public awareness of the university by conducting monthly campus tours and attending national events.

Front-end Developer, 'LikeLion' Developer Union Club

Mar. 2020 – Dec. 2020

- Took courses in HTML, CSS, JavaScript, Django, and Python, and developed the main page front-end for a pdf trading platform webpage.

ADDITIONAL INFORMATION

Certificate: National Accredited ADsP (Advanced Data Analytics Semi-Professional)

Mar. 2022

Programming Skills: Python, R, PyTorch, HTML, CSS, JavaScript, Kotlin, Adobe XD, Figma (all advanced)

Languages: Korean (native fluency), English (full professional proficiency)